

LIST OF MINE SURVEYING PRACTICALS

Course: III Sem. B. Tech (Mining Engineering)

Semester: Monsoon

Sl. No	Aim of experiment	No. of Practical	Learning Outcome
1	Study and sketch of linear measuring instruments (Tapes and Distometer) and carryout linear measurement.	1	Measurement of linear distances using direct and indirect methods
2	Introduction to various angular measurement instruments and measurement of angles using theodolite	1	Measurement of horizontal and vertical angles
3	Closed Traversing using theodolite and distribution of closing errors	1	Transferring coordinates from known baseline
4	Introduction to various Levelling instruments and carryout profile leveling using auto level	1	Transferring of RLs from known benchmark
5	Study and sketch of Total Station and determination of coordinates using Total Station	2	Usage of Total Station and its application in data acquisition
6	Preparation of digital mine plans using Total station data	1	Processing of Total station data and Preparing digital plans
7	Study and sketch of electronic gyro- theodolite (Gyromat-3000) and determination of Gyro-north	1	Obtaining Gyro-meridian and gyro bearing of a survey line using digital gyro theodolite
8	Study and sketch of Nadir Plummet and LASER and demonstration for mine correlation survey	1	Usages and procedure of the modern way of mine correlation survey.
9	Study of GNSS system and carryout GNSS survey in relative point positioning.	1	DGPS surveying for delineation of surface features
10	Study of Terrestrial LIDAR (TLS) and demonstration for Topographical Surveying	1	Application of TLS for surface terrain mapping and modelling